

Preemption' as shown in Figure 3.

The changes are reflected in the `.config` file, as shown below:

```
CONFIG_PREEMPT_VOLUNTARY_BUILD=y
CONFIG_PREEMPT_NONE is not set.
CONFIG_PREEMPT_VOLUNTARY=y
CONFIG_PREEMPT is not set.
CONFIG_PREEMPT_DYNAMIC is not set.
CONFIG_SCHED_CORE is not set.
```

Figure 4 shows kernel configurations for 'Preemptible Kernel' (low latency desktop).

These changes are reflected in `.config` as shown below:

```
CONFIG_PREEMPT_BUILD=y
CONFIG_PREEMPT_NONE is not set.
```

```
CONFIG_PREEMPT_VOLUNTARY is not set.
CONFIG_PREEMPT=y
CONFIG_PREEMPT_COUNT=y
CONFIG_PREEMPTION=y
CONFIG_PREEMPT_DYNAMIC is not set.
CONFIG_SCHED_CORE is not set.
```

Figure 5 shows kernel configurations for 'Fully Preemptible Kernel' (real-time).

These changes are reflected in `.config` as shown below:

```
CONFIG_HAVE_PREEMPT_LAZY=y
CONFIG_PREEMPT_LAZY=y
CONFIG_PREEMPT_NONE is not set.
CONFIG_PREEMPT_VOLUNTARY is not set.
CONFIG_PREEMPT is not set.
CONFIG_PREEMPT_RT=y
```

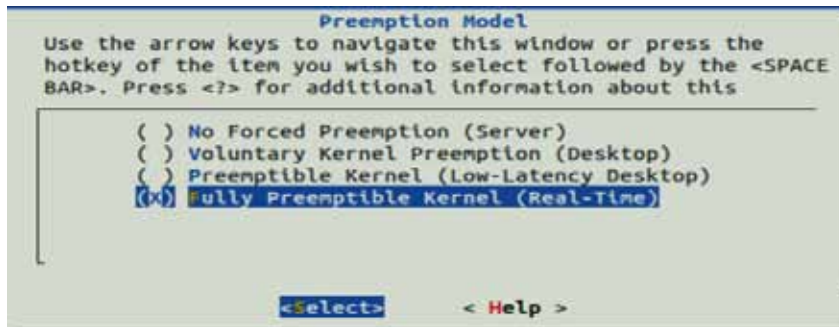


Figure 5: Kernel config options – 'Fully Preemptible Kernel'

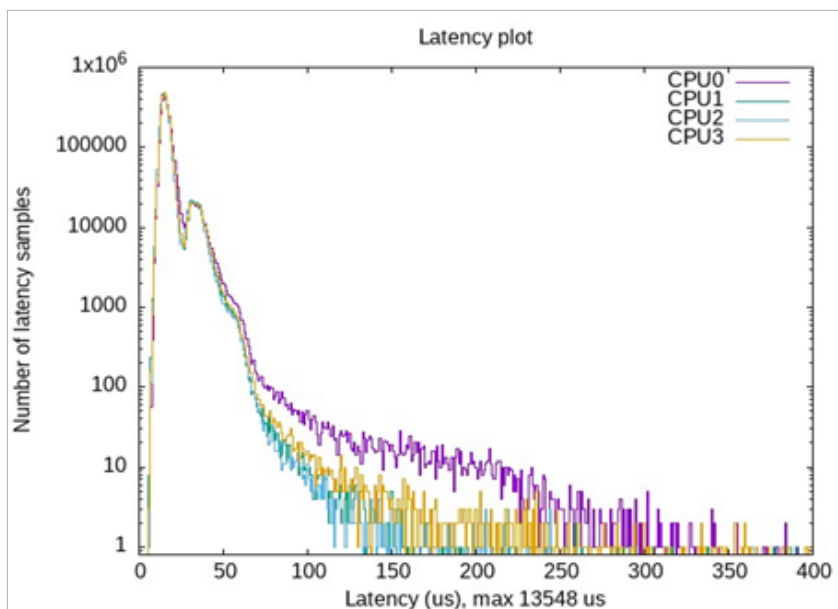


Figure 6: Graph of No Forced Preemption

```
CONFIG_PREEMPT_COUNT=y
CONFIG_PREEMPTION=y
CONFIG_SCHED_CORE is not set.
```

Table 2 lists all kernel config and cyclic test observations.

Figure 6 shows a histogram of No Forced Preemption (server), Figure 7 of Voluntary Kernel Preemption (desktop), Figure 8 of Preemptible Kernel (low latency desktop), and Figure 9 of Fully Preemptible Kernel (real-time).

Figure 10 shows some PREEMPT_RT patch benefits.

Guidelines for using PREEMPT_RT patch in Linux

Kernel guidelines

- **Select a compatible kernel version:** Choose a kernel version that is compatible with the PREEMPT_RT patch we intend to apply.
- **Hardware and driver support:** Ensure that hardware and drivers are supported by the selected kernel version.
- **Patch the kernel:** Apply the PREEMPT_RT patch to the chosen kernel version. Follow the patch's documentation and instructions carefully.
- **Configuration options:** Enable necessary real-time options in the kernel configuration.
- **Compile the source code:** Compile the source code after patching as per the instructions.
- **Real-time scheduling policies:** Configure the kernel to support real-time scheduling policies like SCHED_FIFO and SCHED_RR. These are essential for managing real-time tasks.
- **Minimise interrupt handling latency:** Optimise interrupt handling to minimise latency. Make the interrupt service routines as efficient as possible.
- **Community and support:** Engage with the PREEMPT_RT community and mailing lists for assistance, advice, and troubleshooting.